Date:23.11.2020

2020-2021 FALL SEMESTER HYDRAULICS COURSE

Name: **Surname:** **Student ID:**

Question 3 (25):

In the figure below, the discharge from point D to reservoir C is **Q=0.1bc** m^3/s . Determine the discharge at each pipe and find the elevation of the reservoir A and draw the energy grade line (EGL) and hydraulic grade line (HGL) of the system. ($g=9.81\ m/s^2$).

Important Note: All the lengths and elevations of the reservoirs are in **meter**s. In Figure, L is the length of the pipe, D is the diameter of the pipe, n is Manning friction factor, C is Chezy coefficient, and λ is Darcy-Weisbach friction factor.

Where c,b, and a are the last three digits of your student ID. For example, if the student ID is 1234**567**, **a is 5**, **b is 6** and **c is 7**. For this student ID, for example: L_{AD}=2700 m, D_{AD}= 0.37 m, z_c= 25 m, C=55 $m^{1/2}/s$, n=0.026, λ=0.026 and Q_{DC}=0.167 m^3/s.

